Biochemistry Cumulative Exam
November 13, 2015

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid glutamate, but no carbohydrates, or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (40) Outline a metabolic pathway for the mouse to synthesize alanine on this diet using glutamate as the sole source of carbon, nitrogen and energy. How many glutamate molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of alanine? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, $FADH_2$ and CO_2 generated.

2. (40) Outline a metabolic pathway for the mouse to synthesize isocitrate on this diet using glutamate as the sole source of carbon, and energy? How many glutamate molecules must be used to provide enough carbon atoms, ATP, and NADH molecules to synthesize each molecule of isocitrate? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, $FADH_2$ and CO_2 generated.

2. (20) If glutamate is labeled with ^{14}C at carbon 3, which carbon(s) of alanine and isocitrate will first become labeled?

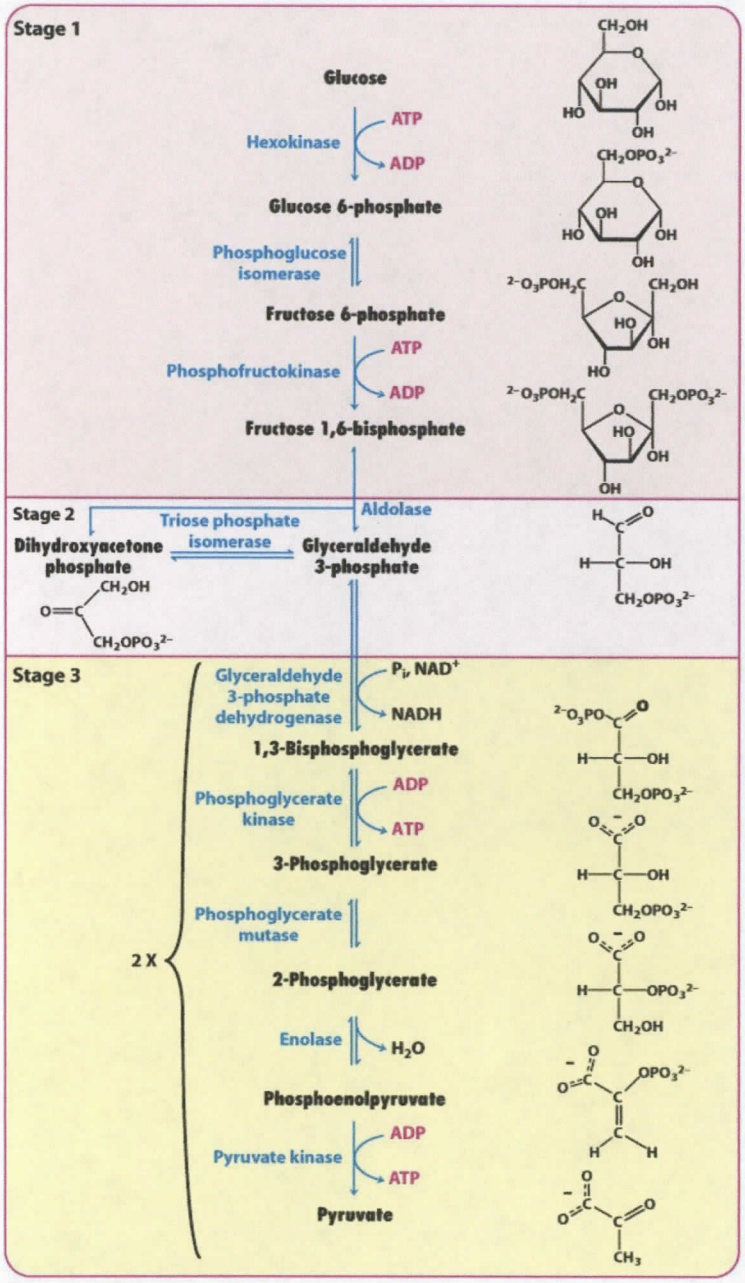

Figure 16-2
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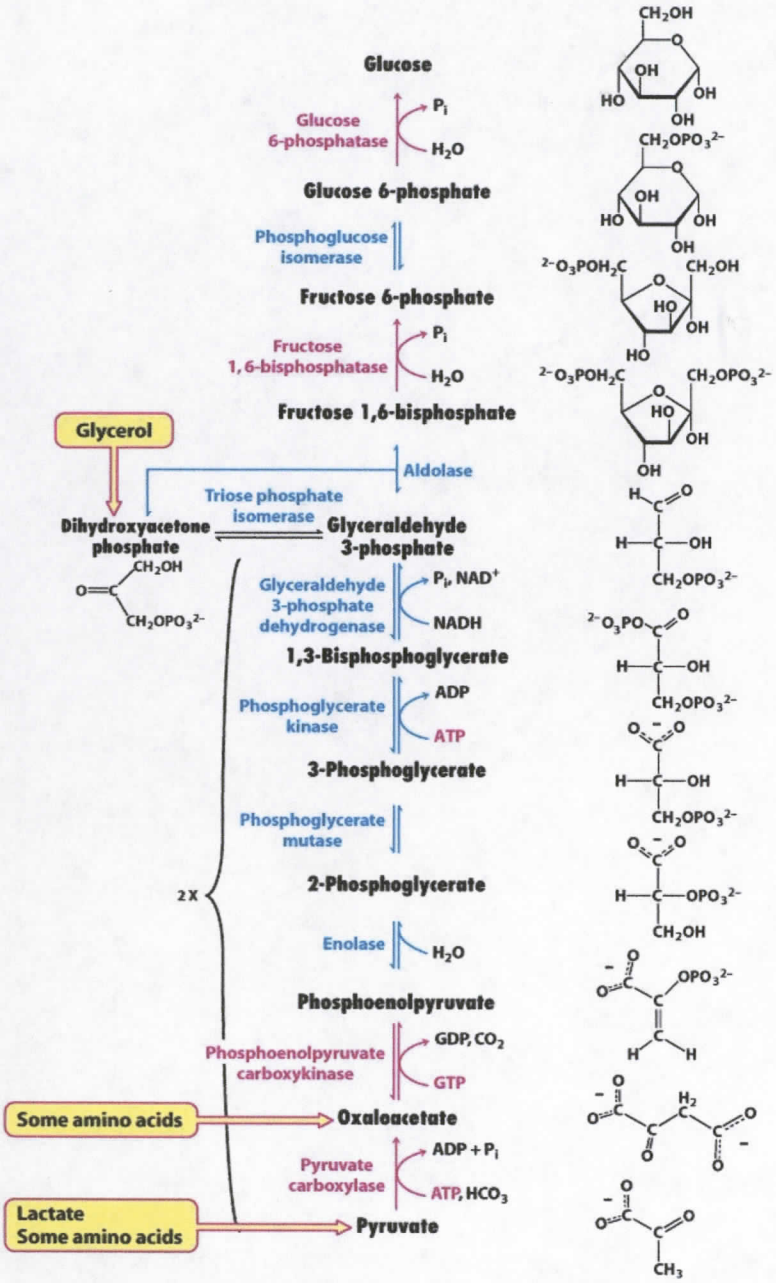

Figure 16-22
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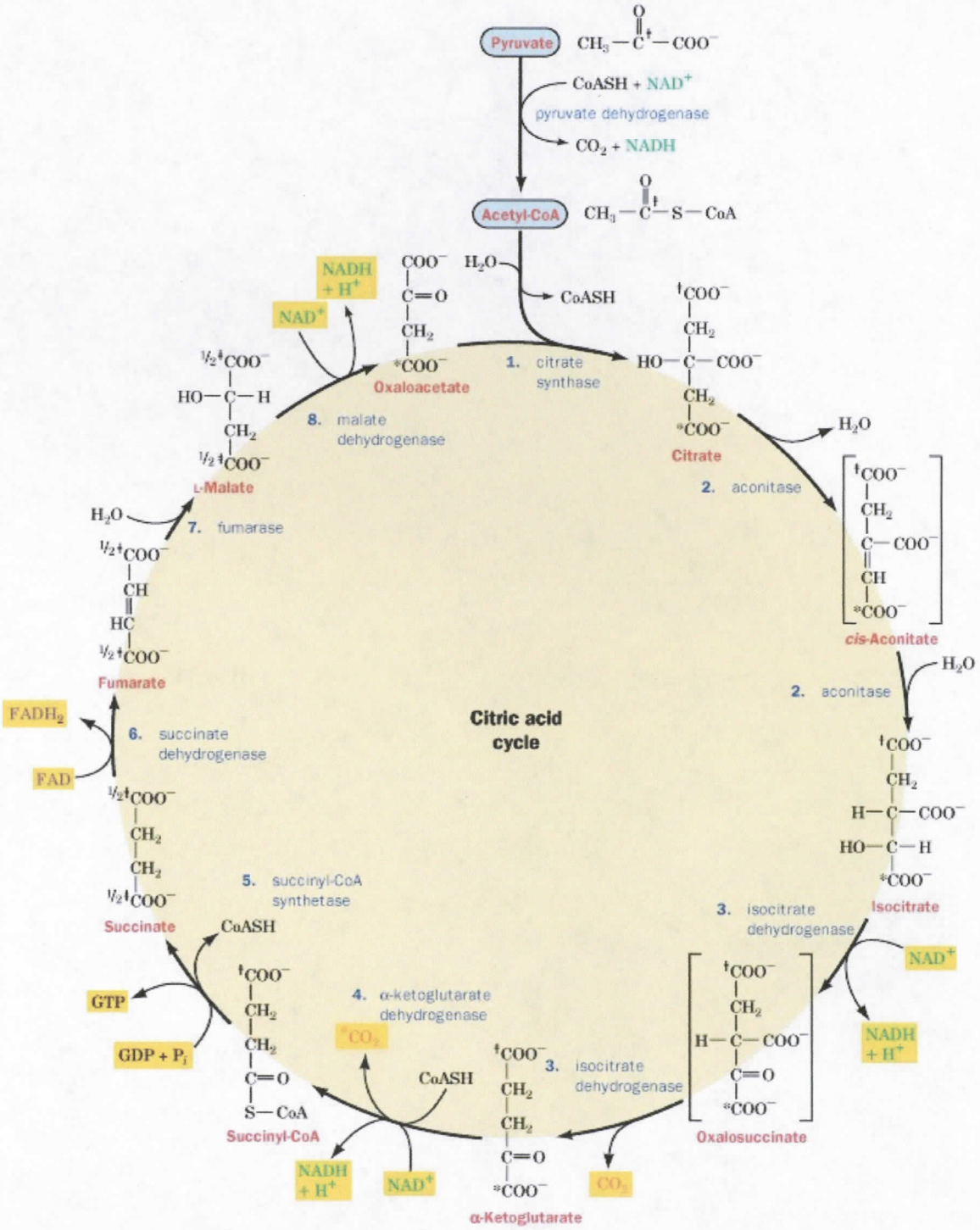
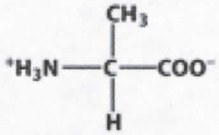
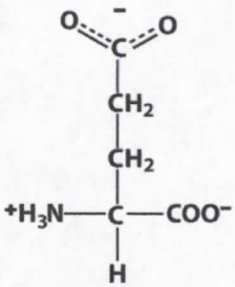
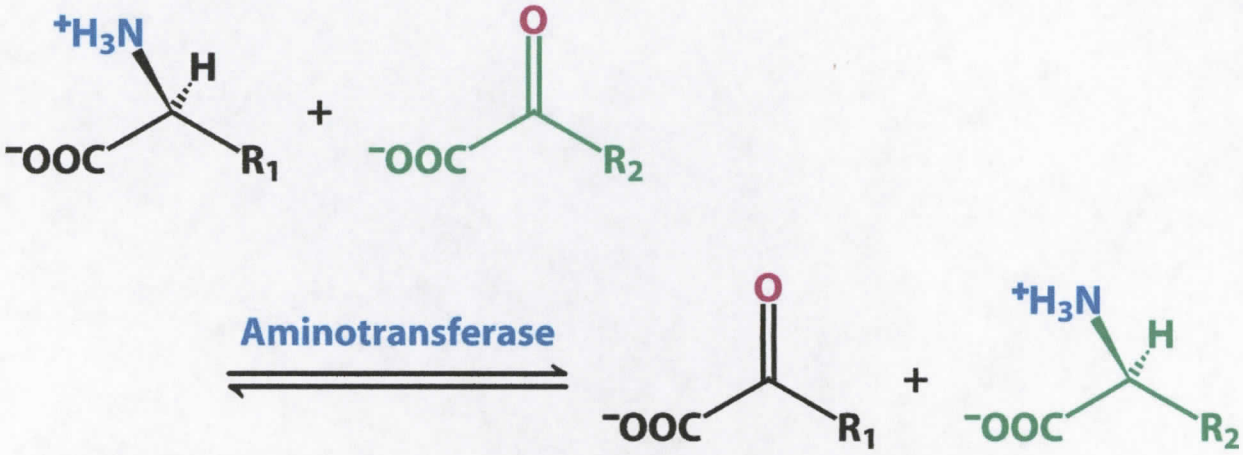

Pyruvate
CoASH + NAD+
pyruvate dehydrogenase
CO2 + NADH
Acetyl-CoA
H2O
CoASH
1. citrate synthase
Oxaloacetate
Citrate
2. aconitase
H2O
cis-Aconitate
2. aconitase
H2O
Isocitrate
3. isocitrate dehydrogenase
NAD+
NADH + H+
Oxalosuccinate
3. isocitrate dehydrogenase
CO2
α-Ketoglutarate
4. α-ketoglutarate dehydrogenase
CO2
CoASH
NADH + H+
NAD+
Succinyl-CoA
5. succinyl-CoA synthetase
CoASH
GTP
GDP + Pi
Succinate
6. succinate dehydrogenase
FADH2
FAD
Fumarate
7. fumarase
H2O
L-Malate
8. malate dehydrogenase
NADH + H+
NAD+
Citric acid cycle



Alanine
(Ala, A)



Glutamate
(Glu, E)



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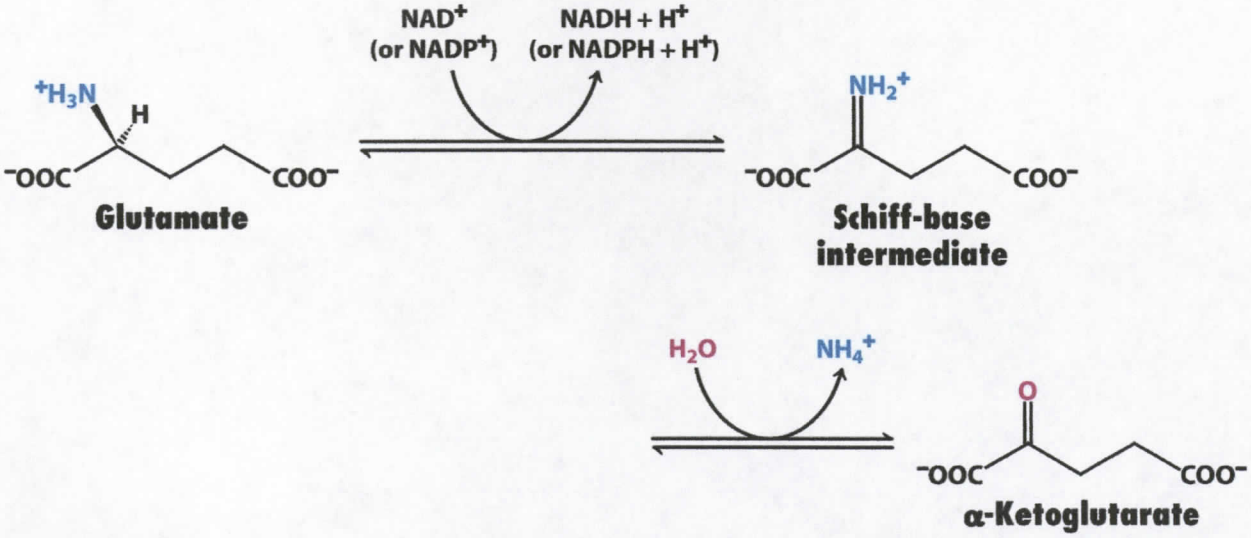

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